

PLANNED EVENT MANAGEMENT SYSTEM

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1. Introduction

1.1 Background

Sri Lanka Telecom (SLT) manages all the new connection requirements using an operational support system called "Clarity". When providing fiber related product/service is necessary to ensure the fiber availability within the required area. However, there are areas where the fiber laying is not done or in progress under the network planning project. Clarity system is used to monitor such projects and monitor the newly required fiber laying project progress. There are defined tasks assigned to multiple teams to fulfill the fiber laying task within agreed time frame and each team has its own operational level agreements.

However, coordinating tasks across multiple teams lacks real-time visibility due to lack of structured communication, inefficient task tracking, and other external factors like customer side dependencies and issues. Therefore, to enhance efficiency, we propose a Planned Event Management System that centralizes task tracking, automates notifications, and ensures timely completion of tasks within Operational Level Agreements (OLAs). This system will improve workflow transparency, accountability, and overall service quality.

1.2 Overview

The proposed system aims to address the inefficiencies in SLT's fiber connection installation process by providing real-time visibility, structured communication, and accountability mechanisms. The system will introduce a centralized dashboard with search functionalities, task tracking, priority handling, and automated notifications triggered when Operational Level Agreements (OLAs) exceed and send escalations for delayed tasks and automatically adjust the timeline. By improving end-to-end tracking, the system ensures timely completion of tasks, reduces delays, and enhances customer satisfaction.

1.3 Proposal

To overcome the existing challenges, we propose a new system with the following key features:

- **PE Management** – Track and update PE status.
- **User & Workgroup Management** – Assign roles and manage teams.
- **Task Management** – Monitor tasks, escalations, and make tasks urgent
- **Notifications** – Send OLA reminders and escalate overdue tasks.

- **Dashboard** – Search, filter, and track planned events.
- **Timeline management** - Update timeline dynamically.
- **Issue Reporting** - Contact relevant workgroups if an issue arises.

1.4 Objectives

1. Enhancing Efficiency.
2. Ensuring Timely Completion.
3. Increasing Accountability.
4. Improving Customer Satisfaction.
5. Improving communication and visibility.

1.5 Scope

The SFC Dashboard System streamlines fiber connection management by tracking Planned Events (PE) and task progress in real time. It enhances accountability with profile-based access, allowing admin to manage teams while ensuring seamless communication through notifications and escalations. The dashboard enables efficient search and filtering of PEs, while the Super Admin can modify tasks, workgroups, and OLA settings to maintain flexibility and efficiency.

2. Stakeholders

- **Project Supervisor:** Mrs.Yamuna Subramaniam
- **Development Team:** SLT Digital Platform (Interns of the SFC Dashboard Project)
- **End Users:** Admin, General Manager, Account Manager, Engineer-NET-PROJ-ACC-CABLE, Engineer-NET-PLAN-TX, Engineer-Service-Fulfillment, Network Engineer, Engineer-NET-PLAN-ACC, OPMC Manager, Staff member

3. Functional Non-Functional Requirements

3.1 Functional requirements

3.1.1 Search PE

The system should allow users to search for Planned Events (PE) using PE number, job reference number, SO number, region, area, task, work group, network engineer, or customer, and view the current status, assigned workgroup, and duration of the PE.

3.1.2 View dashboard

The dashboard should display the number of urgent, in progress, OLA validation, and hold tasks for the user and get details of them.

3.1.3. Assign priority

The system should allow priority marking for urgent PE processes. The Account manager can request to mark a PE process as urgent if the general manager accepts it and the PE should go to the urgent list.

The General Manager should be able to mark a task as urgent for a specific workgroup, moving it to the urgent task list from either the "In Progress" or "OLA Violated" list. This urgency status remains until the task is closed. However, if the task moves to the next workgroup, it will no longer be marked as urgent.

3.1.4. Send escalation

Escalation mail should be sent to relevant stakeholders if a task is not completed on time.

3.1.5. Auto-adjust timeline and hold task

Task durations should be auto-adjusted based on whether tasks are completed earlier or later than the scheduled time or deadline extent requests.

When a task goes to a process that doesn't have an OLA the duration calculation must freeze and it is added to hold tasks list after coming back to normal flow the duration must start to calculate and the time taken to the side process must be added to duration.

3.1.6. Admin privileges and system management

Admin should have full access to the system and can add or remove tasks,workgroups. Also they should be able to add and remove members from workgroups.

3.1.7. Report issues

Users should be able to report issues and transfer tasks to other work groups and also can send a request to extend the deadline. If the account manager accepts the request, the deadline should be extended.

3.1.8. Update the estimated duration of an assigned task

Engineer-NET-PROJ-ACC-CABLE should be able to update the estimated duration of an assigned task. The updated duration can be viewed by everyone in the system.

3.1.9. Assign tasks

OPMC manager should be able to assign tasks to workgroups and individuals. The relevant workgroups will be notified.

3.2 Non-Functional Requirements

3.2.1 Security

- The system should implement role-based access control (RBAC) to restrict functionalities based on user roles.
- All communications should be encrypted using industry-standard protocols.
- User authentication should include multi-factor authentication (MFA) to prevent unauthorized access.
- Audit logs should be maintained to track system access and modifications.
- Data backups should be performed regularly to prevent data loss.

3.2.2 Usability

- The system should have an intuitive user interface with easy navigation.
- The dashboard should provide a clear visual representation of PE statuses.
- The system should support accessibility features for users with disabilities.
- Notifications should be clear and actionable, ensuring users respond in a timely manner.

3.2.3 Scalability

- The system should be designed to handle increasing workloads as more PEs are processed.
- The architecture should support the addition of new features and integrations without major rework.
- The database should be optimized for efficient querying and large data volumes.
- The system should support multiple workgroups working concurrently without performance degradation

4. User Personas

- Admin
- General Manager
- Account Manager
- Engineer-NET-PROJ-ACC-CABLE

- Engineer-NET-PLAN-TX
- Engineer-Service-Fulfillment
- Network Engineer
- Engineer-NET-PLAN-ACC
- OPMC Manager
- Staff member

4.1 User Personal Characteristics

Admin

- Log in
- Add and remove tasks, workgroups.
- Edit task details (OLA)
- Search PE and see where the PE is and which group works for that.
- View the number of tasks and see their details.
- Assign members to the workgroup.

General manager

- Log in.
- Make tasks as urgent and accept urgent task requests.
- Search and filter based on PE number, Region, Area, task, work group network engineer or Customer to see more information.
- View the number of tasks and see their details.

Account Manager

- Log in.
- Request urgent PE process.
- Search and filter based on PE number or Customer to see more information.
- View the number of tasks assigned for them.

Engineer-NET-PROJ-ACC-CABLE

- Log in.
- Update the estimated duration of an assigned task.
- Search and filter based on PE number, job reference number, SO number, Region, network engineer or Customer to see more information.
- View the number of tasks assigned for each type.
- Report issues to other work groups.

Engineer-NET-PLAN-TX

- Log in.

- Search and filter based on PE number, SO number to see more information.
- View the number of tasks assigned for each type.
- Report issues to other work groups.

Engineer-Service-Fulfillment

- Log in.
- Search and filter based on PE number, job reference number, SO number, Region, or Customer to see more information.
- View the number of tasks assigned for each type (Urgent, in progress, OLA Validation, Hold).
- View total number of PE in the system.

Network Engineer

- Log in.
- Search and filter based on PE number, SO number to see more information.
- View the number of tasks assigned for each type.
- Report issues to other work groups.

Engineer-NET-PLAN-ACC

- Log in.
- Search and filter based on PE number, SO number to see more information.
- View the number of tasks assigned for each type.
- Report issues to other work groups.

OPMC Manager

- Log in.
- Search and filter based on PE number, SO number to see more information.
- View the number of tasks assigned for each type.
- Assign tasks to workgroups.
- Report issues to other work groups.

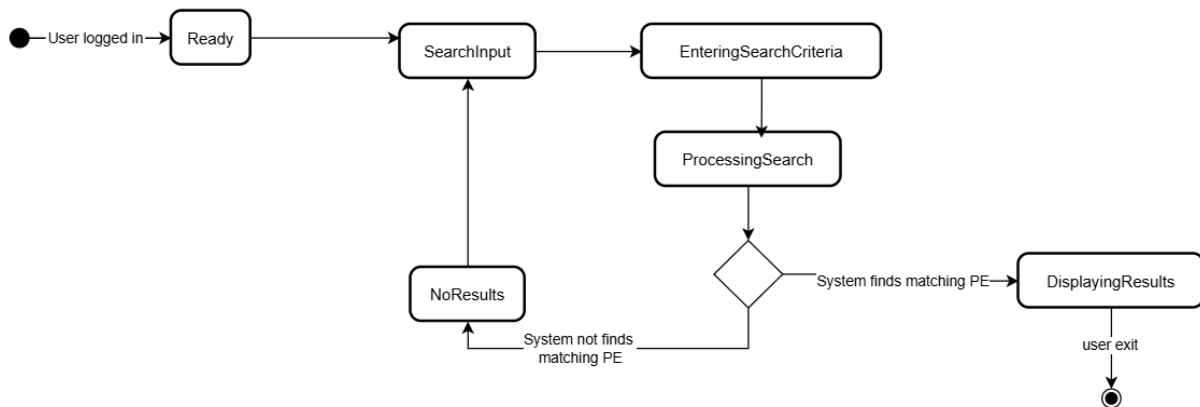
Staff member

- Log in
- View the number of tasks assigned to their workgroup and see task details.
- View the auto-adjusting PE duration.

5. Use cases for the SFC Dashboard project

Actors : Admin, General Manager, Account Manager,
 Engineer-NET-PROJ-ACC-CABLE, Engineer-NET-PLAN-TX,
 Engineer-Service-Fulfillment, Network Engineer, Engineer-NET-PLAN-ACC, OPMC
 Manager, Staff member

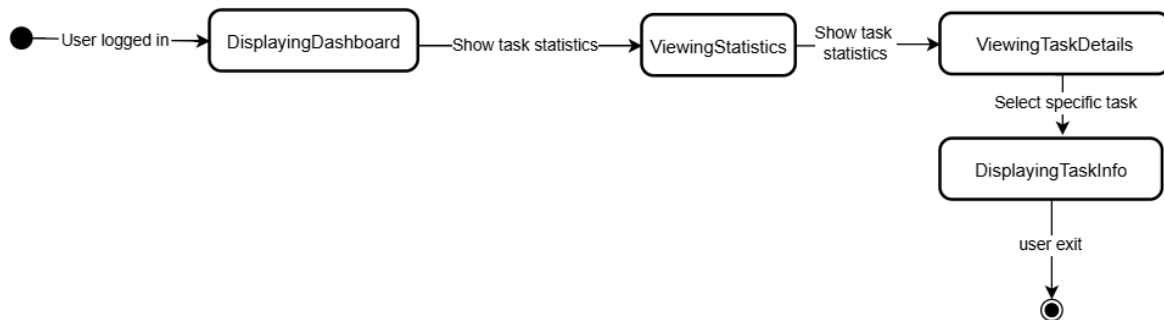
Use case 1: Search for Planned Events (PE)



Actor	All users
Preconditions	User is logged into the system. User has the required permissions.
Flow of Steps	1. The user enters search criteria (PE number, job reference number, SO number, region, area, task, workgroup, network engineer, or customer). 2. The system retrieves matching PEs. 3. The system displays the PE details, including current status, assigned workgroup, and timeline.
Postconditions	The user successfully views the relevant PE details.

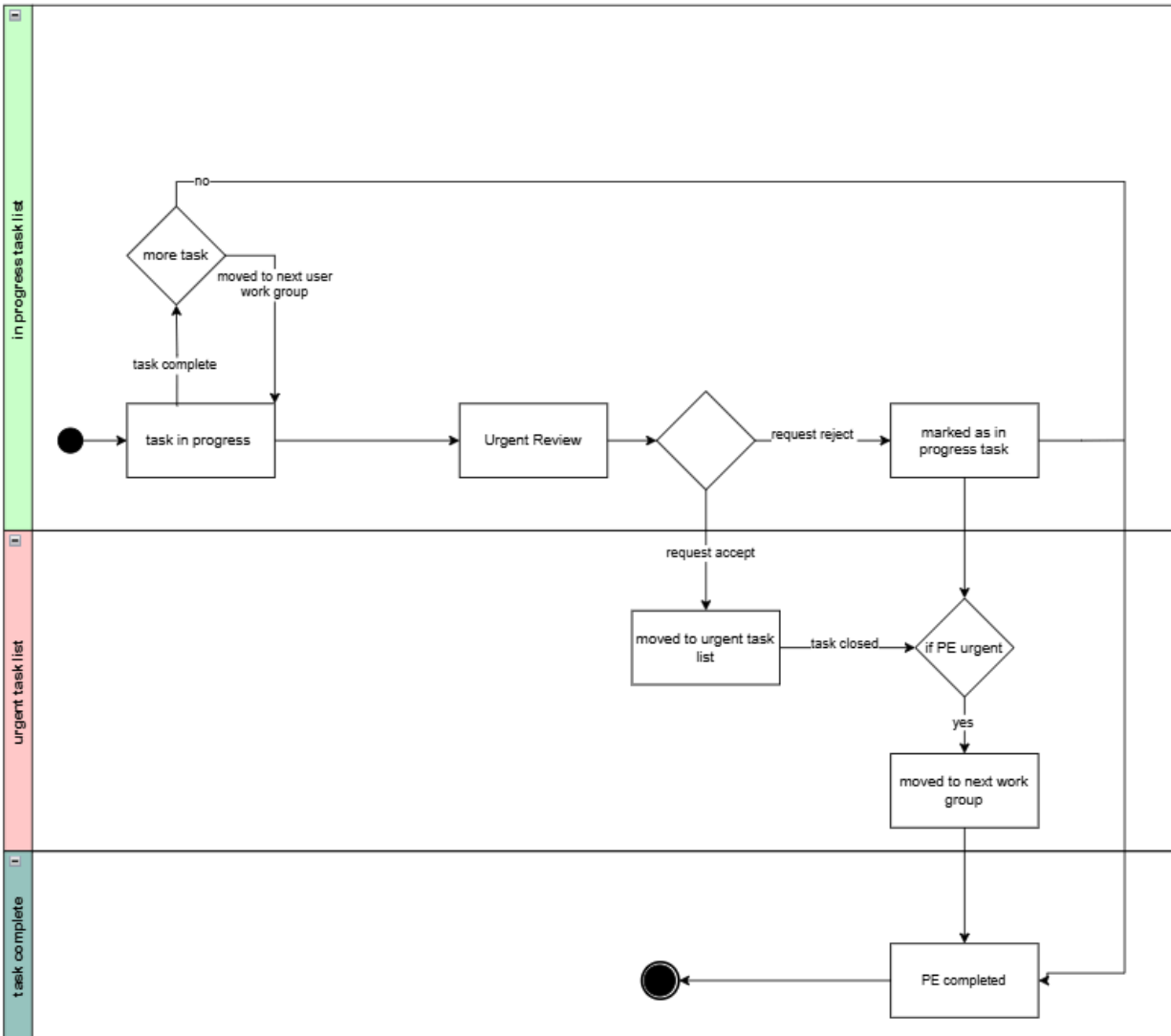
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Use case 2: View Dashboard Statistics



Actor	All users
Preconditions	User is logged into the system.
Flow of Steps	1. The user navigates to the dashboard. 2. The system displays the number of urgent, in-progress, OLA validation, and hold tasks. 3. The user clicks on a category to see task details.
Postconditions	The user successfully views categorized tasks and details.

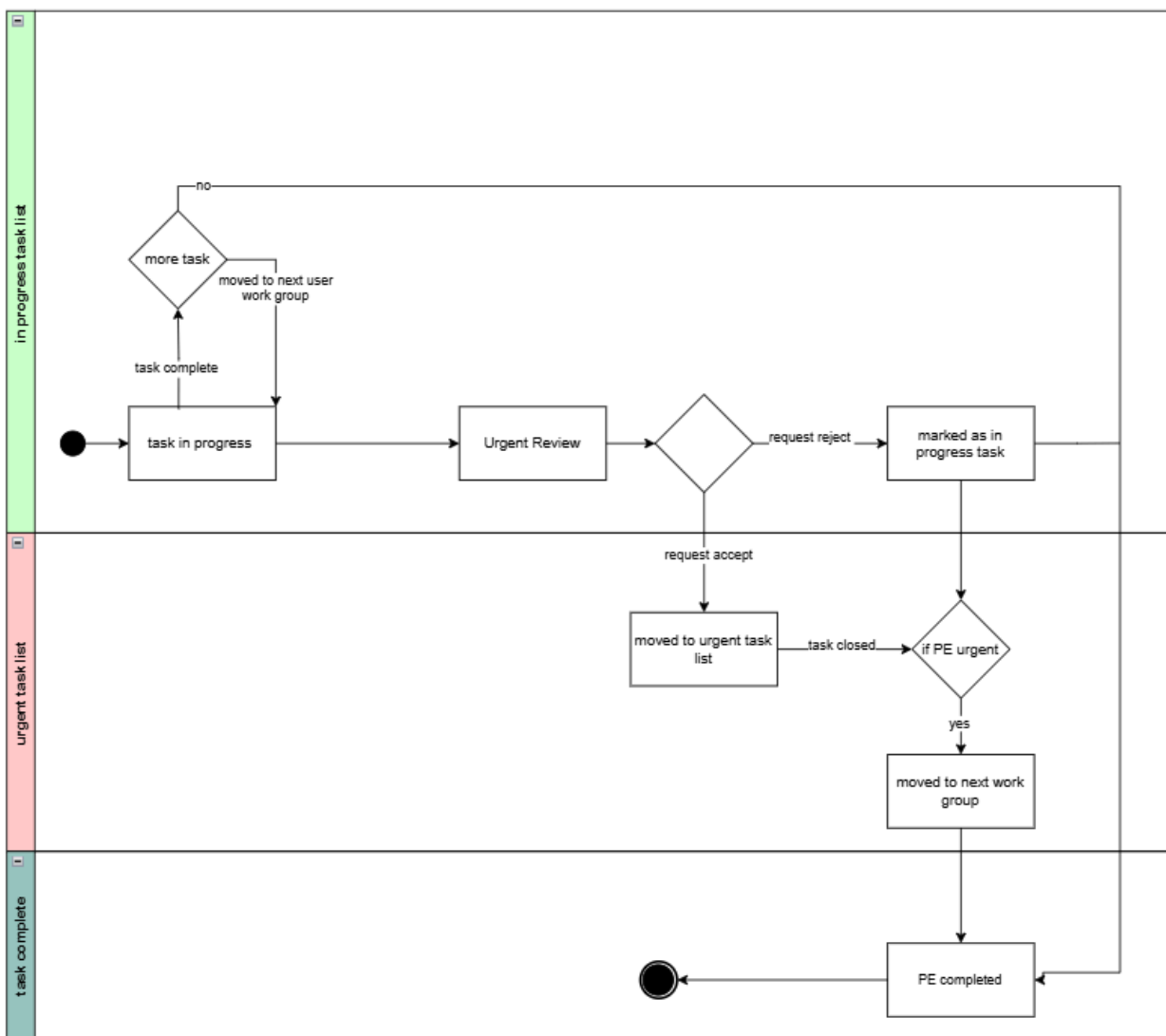
Use case 3: Request to Mark a PE Process as Urgent



Actor	Account Manager
Preconditions	Account Manager is logged in.
Flow of Steps	1. The Account Manager selects a PE. 2. The Account Manager marks the PE as urgent. 3. The system updates the PE status and notifies relevant

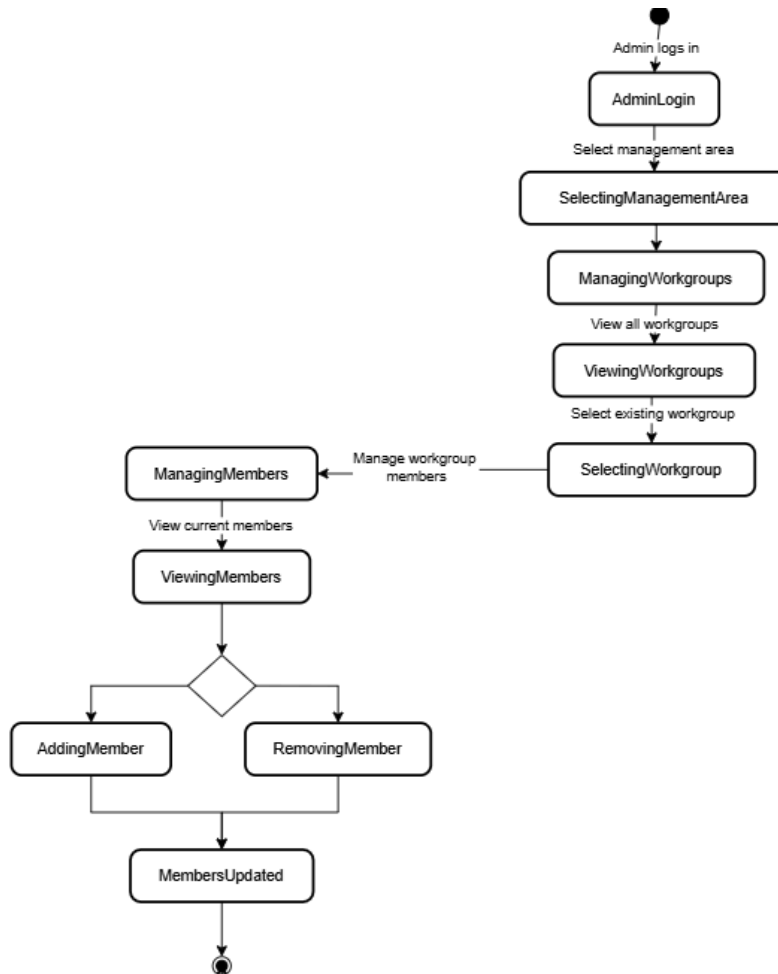
	stakeholders.
Postconditions	The PE is sent to the service fulfillment group to confirm as an urgent PE.

Use case 4: Confirm Urgent PE process



Actor	General Manager
Preconditions	General manager is logged in. The system has urgent PE pending confirmation.
Flow of Steps	1. The system notifies the general manager about newly marked urgent PE. 2. General manager reviews and confirms urgency. 3. The system updates the PE status.
Postconditions	Urgent PE are confirmed and updated.

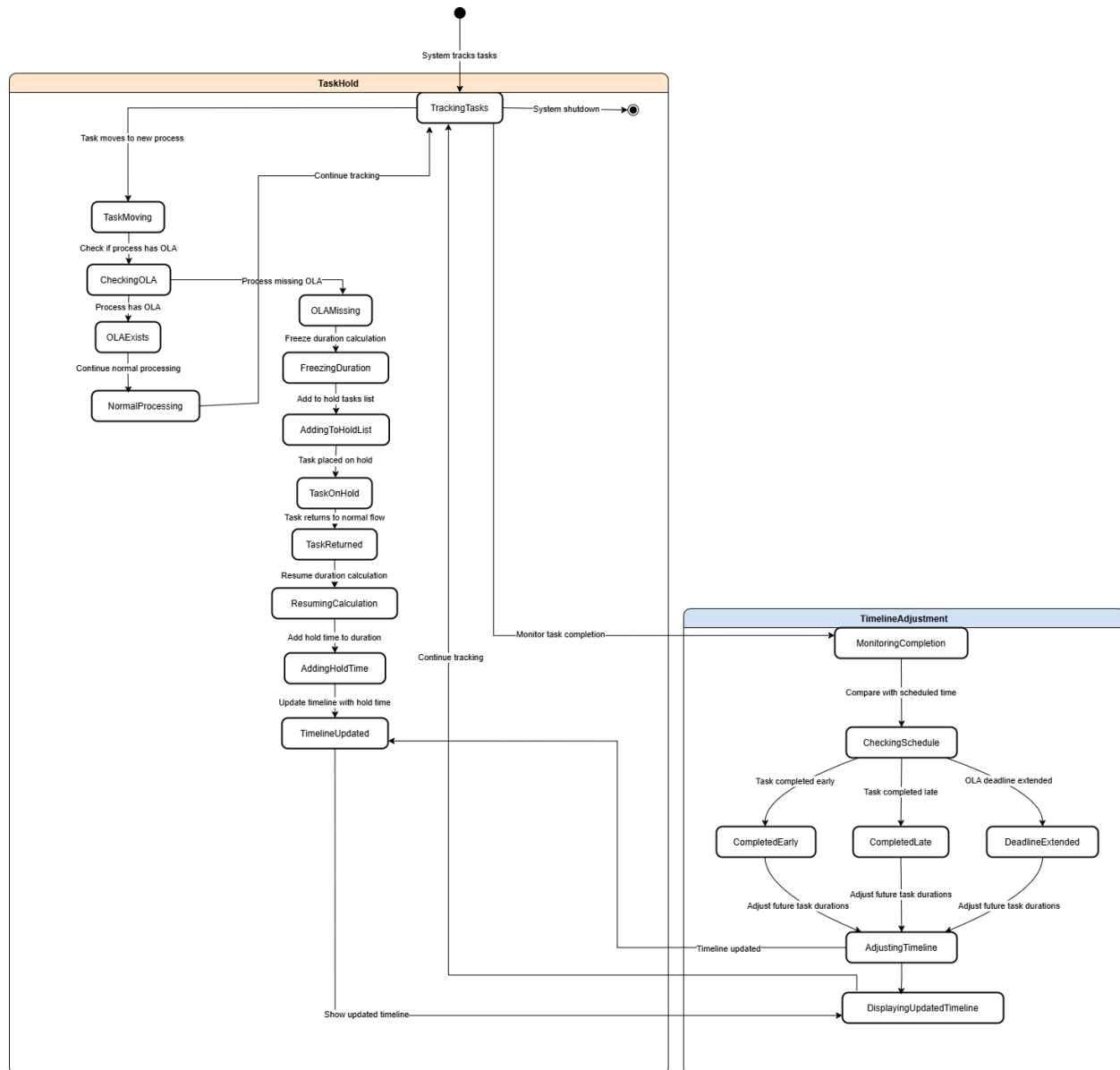
Use case 5: Manage Staff members



Actor	Admin
Preconditions	Admin is logged in.
Flow of Steps	1. Admin navigates to workgroup management. 2. Admin selects a workgroup and adds/removes members. 3. The system updates the Staff member list.

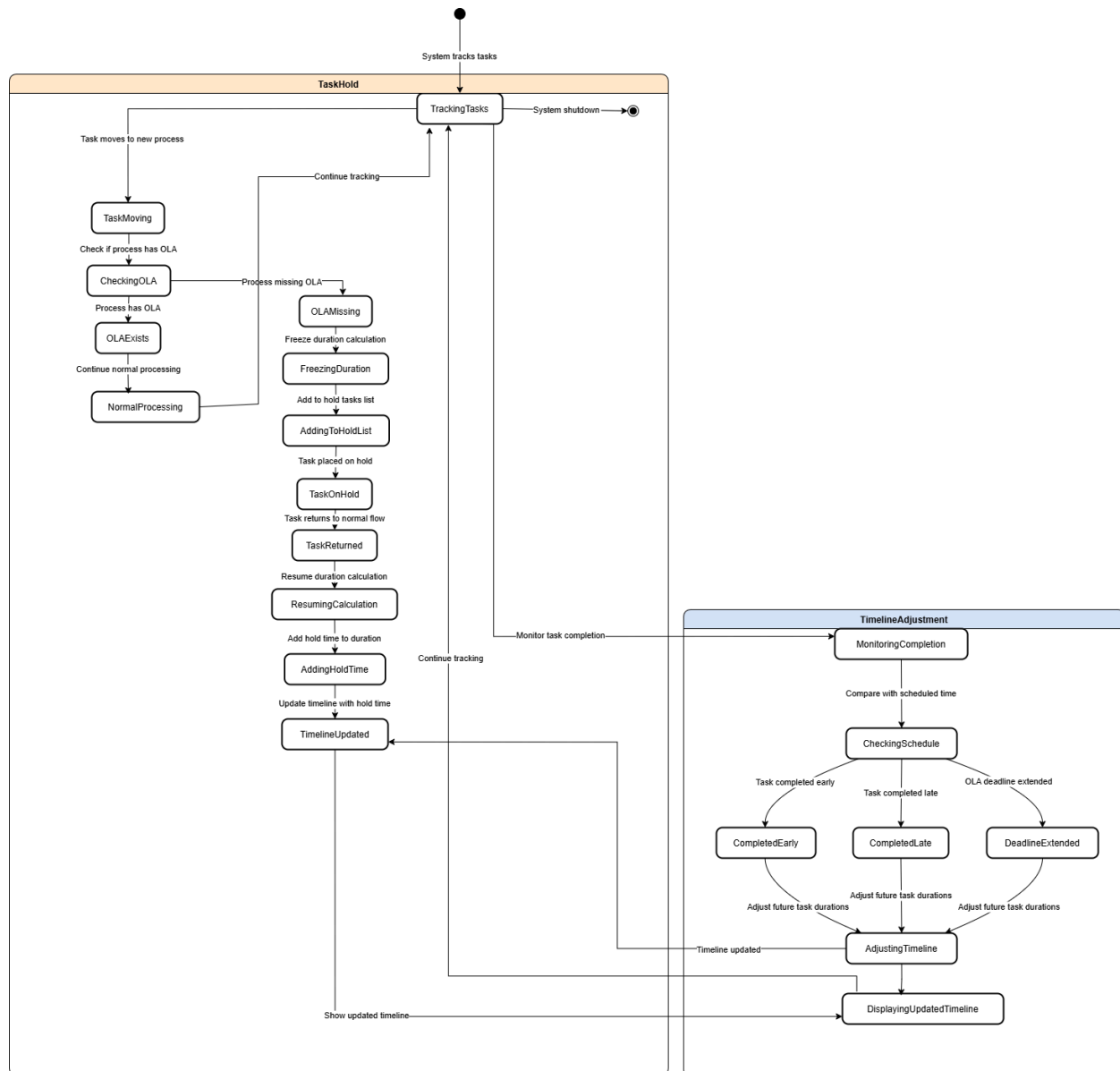
Postconditions	The Staff membership is successfully updated.
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Use case 6: Auto-Adjust Task Duration



Actor	System and All users
Preconditions	The system is tracking task durations.
Flow of Step	1. The system tracks task completion times. 2. If a task is completed earlier or later, the system adjusts future task durations. 3. Users view the updated durations.
Postconditions	Task durations are dynamically adjusted based on actual completion.

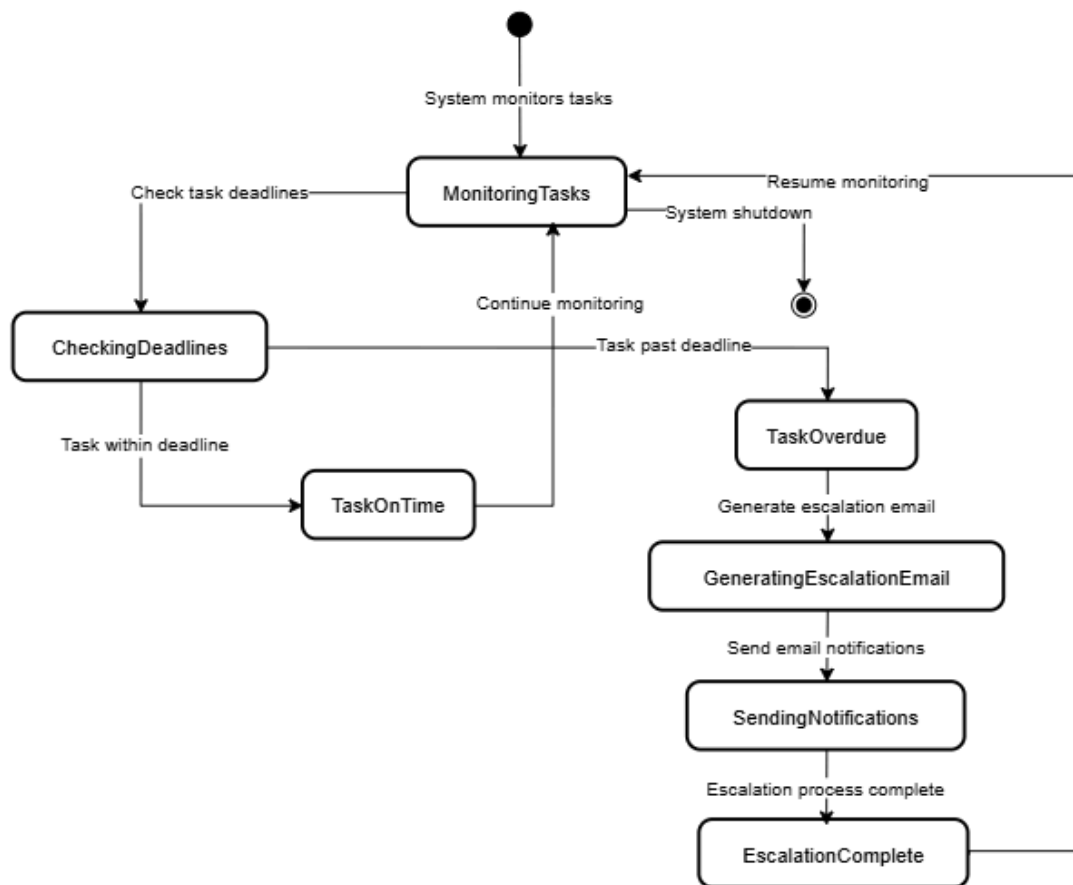
Use case 7: Handle OLA Missing Process and Task Hold



Actor	System
Preconditions	A task moves to a process that lacks OLA.
Flow of Step	1. The system detects a process lacking OLA. 2. The system freezes task duration calculation.

	- The task is added to the "Hold Tasks" list. - When the task returns to a normal process, duration calculation resumes, including hold time.
Postconditions	The system correctly tracks task durations and holds.

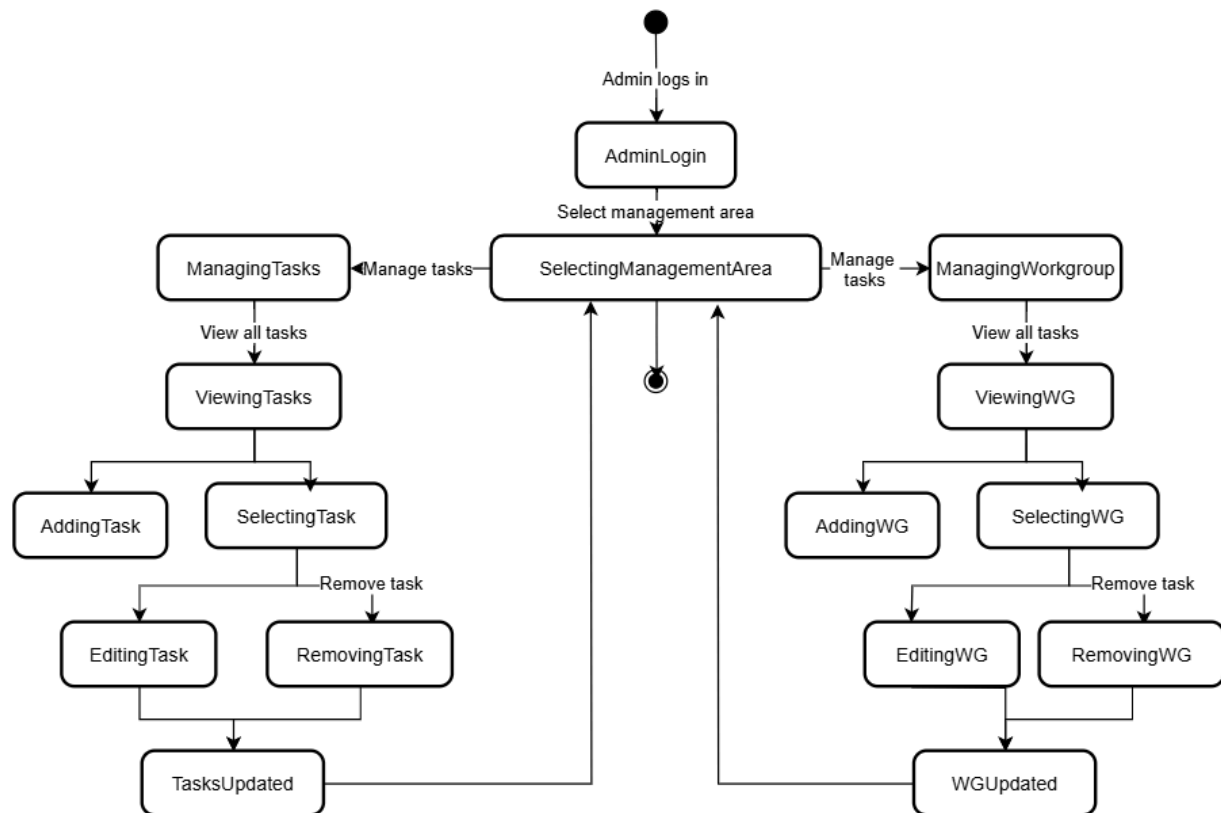
Use case 8: Send Escalation Notification for Overdue Tasks



Actor	System
Preconditions	A task has exceeded its scheduled duration.

Flow of Step	1. The system monitors task deadlines. 2. If a task is overdue, the system sends an escalation notification to the responsible workgroup and admin. 3. The system updates the task status.
Postconditions	Stakeholders are notified of the overdue task.

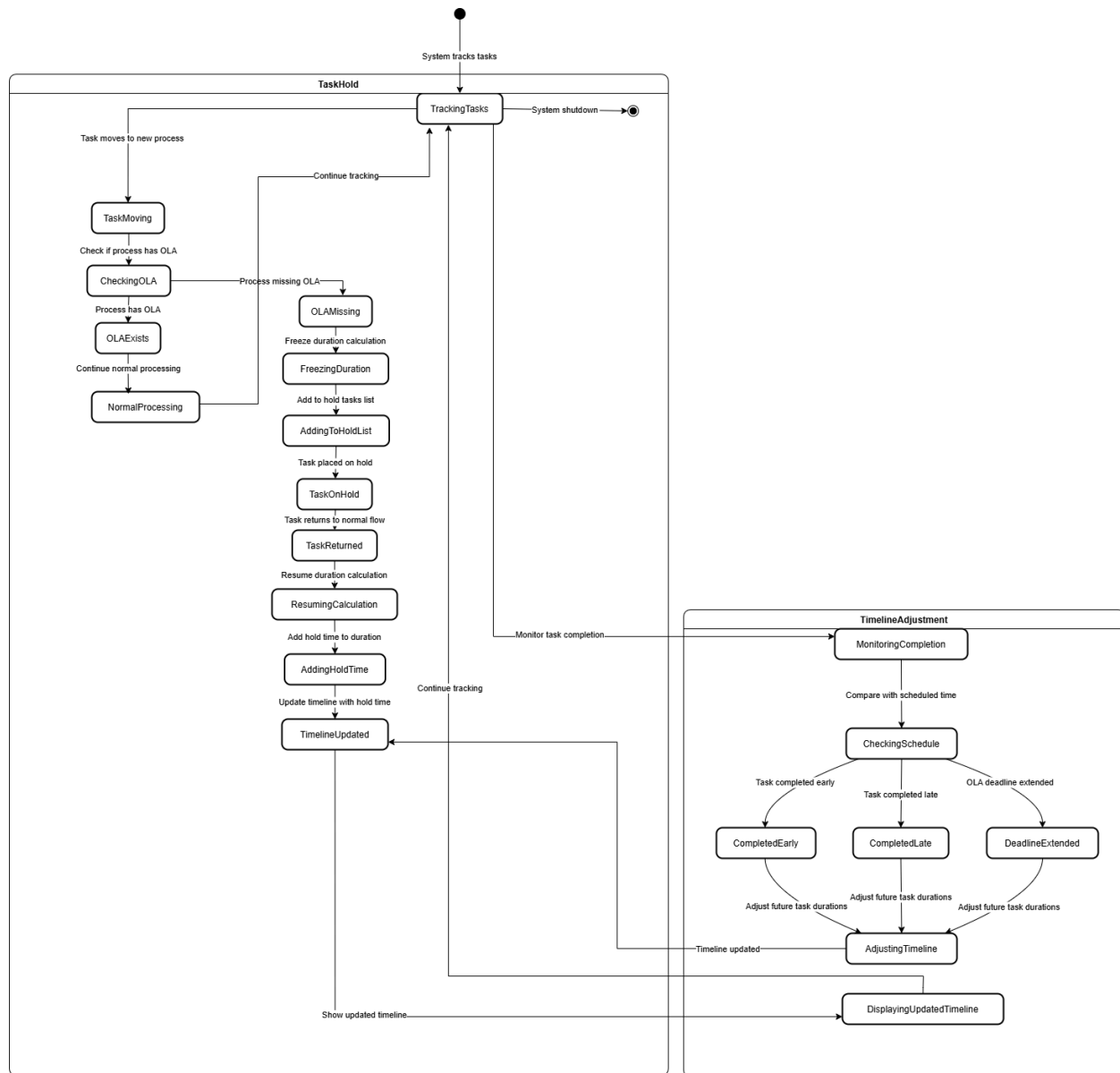
Use case 9: Management of Tasks and Workgroups



Actor	Admin
Preconditions	Admin is logged in.

Flow of Step	1. Admin navigates to the management section. 2. Admin selects and modifies tasks and workgroups 3. Admin edits task details, including OLA settings. 4. The system reflects changes.
Postconditions	Admin successfully manages system entities.

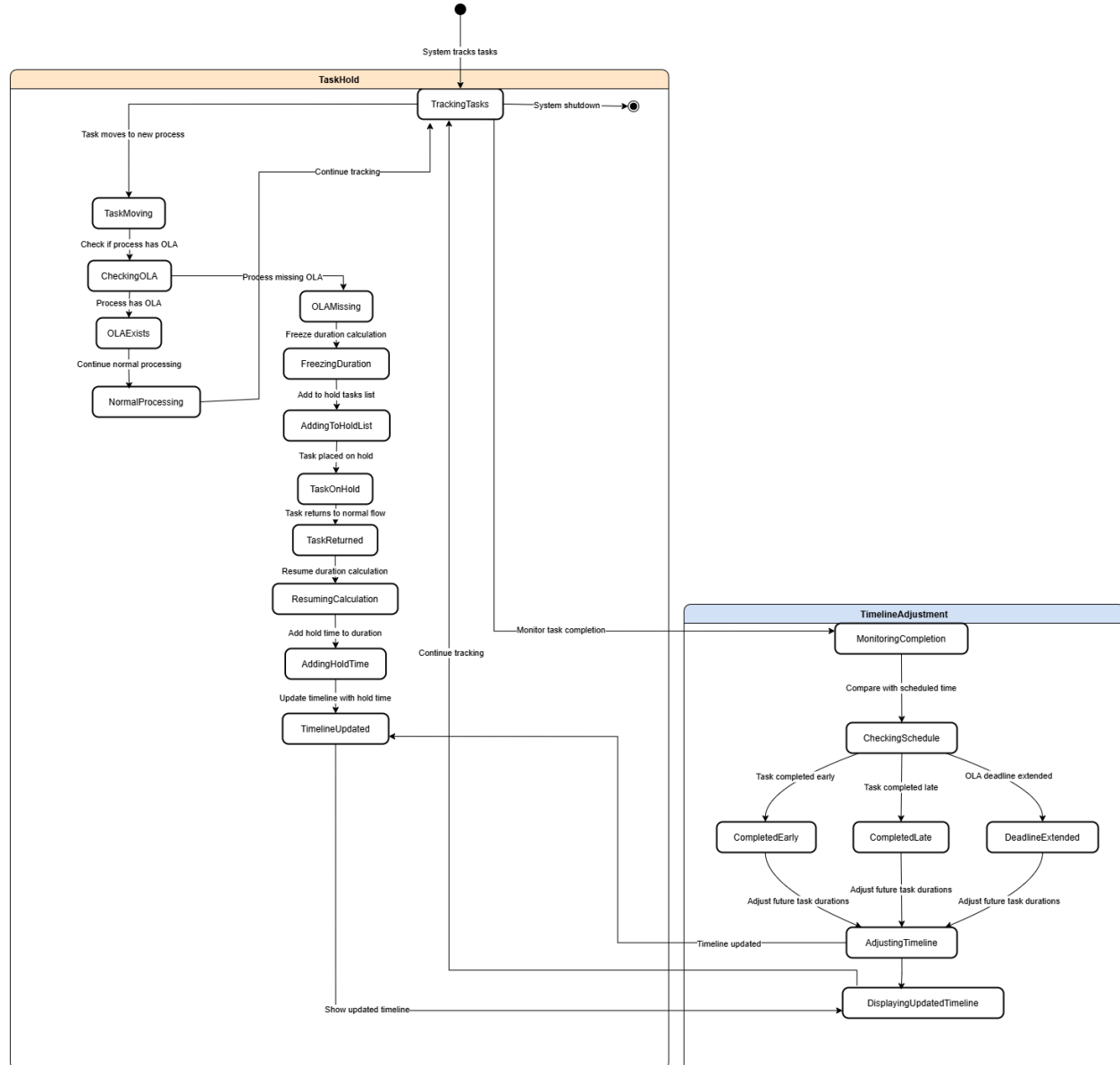
Use case 10: View Auto-Adjusting PE Duration



Actor	All users
Preconditions	System is tracking PE durations.

Flow of Step	1. The system calculates PE durations based on task completion times. 2. The user navigates to task details. 3. The system displays adjusted durations.
Postconditions	The user views the real-time PE duration updates.

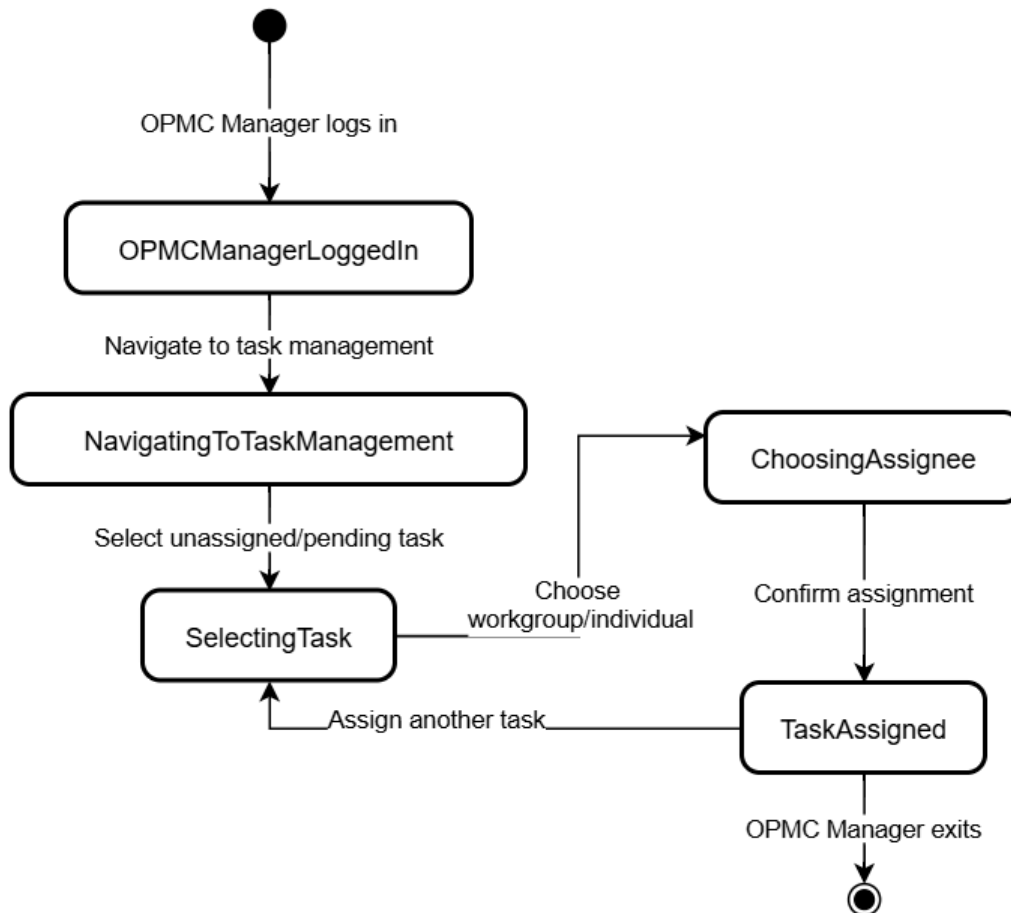
Use case 11: Request OLA Extension



Actor	Staff member, Account manager
Preconditions	Staff members are logged in. Task is nearing or has exceeded its OLA deadline.

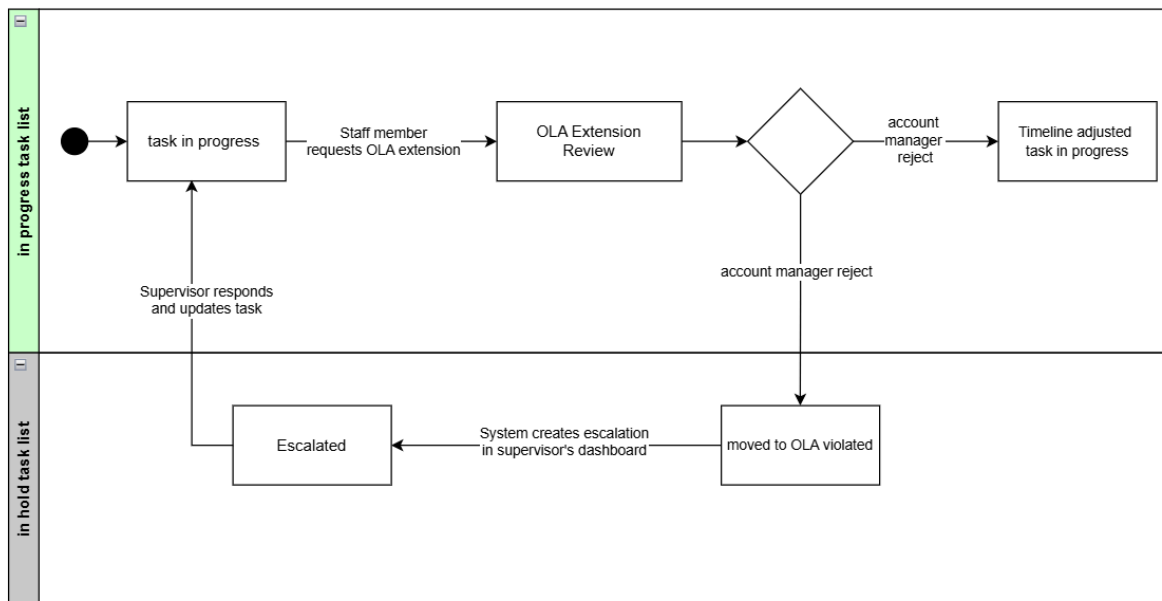
Flow of Step	1. Staff member selects the task and submits an OLA extension request. 2. They provide a reason and suggest a new deadline. 3. System sends the request to the Account Manager for approval. 4. The Account Manager reviews and approves or rejects the request.
Postconditions	If approved, the task's OLA is extended. If rejected, the Staff member is informed.

Use case 12: Assign Task



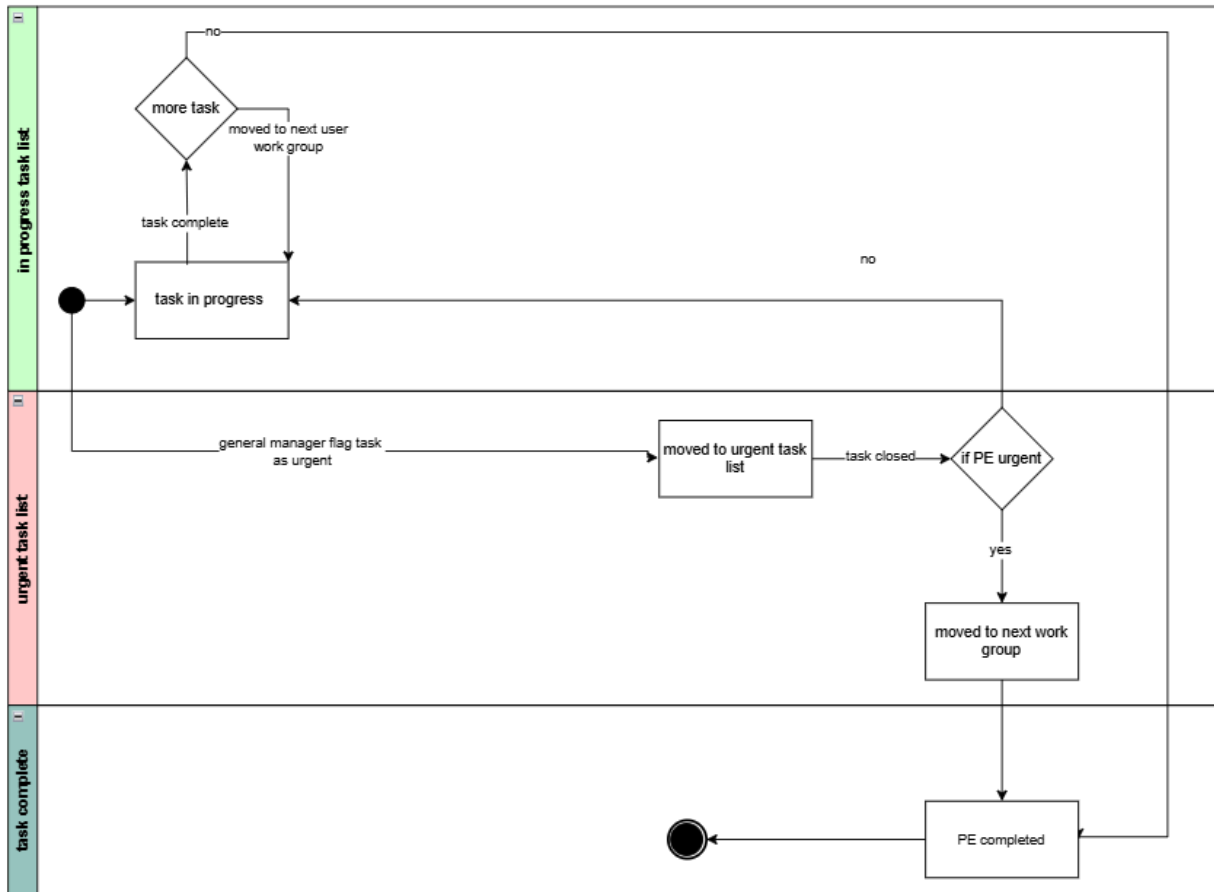
Actor	OPMC Manager
Preconditions	The OPMC Manager is logged into the system. Tasks are available for assignment.
Flow of Step	1. The OPMC Manager navigates to the task management section. 2. Selects an unassigned or pending task. 3. Chooses a workgroup or individual to assign the task. 4. Confirms the assignment. 5. The system updates the task status and notifies the assigned user/workgroup.
Postconditions	The task is successfully assigned. Assigned users receive notifications.

Use case 13: Report issues to other work groups



Actor	Engineer-NET-PROJ-ACC-CABLE, Engineer-NET-PLAN-TX, Network Engineer, Engineer-NET-PLAN-ACC, OPMC Manager
Preconditions	The user is logged into the system. The user has permission to report issues to other work groups. The PE is in progress.
Flow of Step	1. The user navigates to the view details section. 2. Add issue details. 3. Submit the report to the relevant group.
Postconditions	The issue is forwarded to the relevant workgroup. PE status is updated to “on hold”.

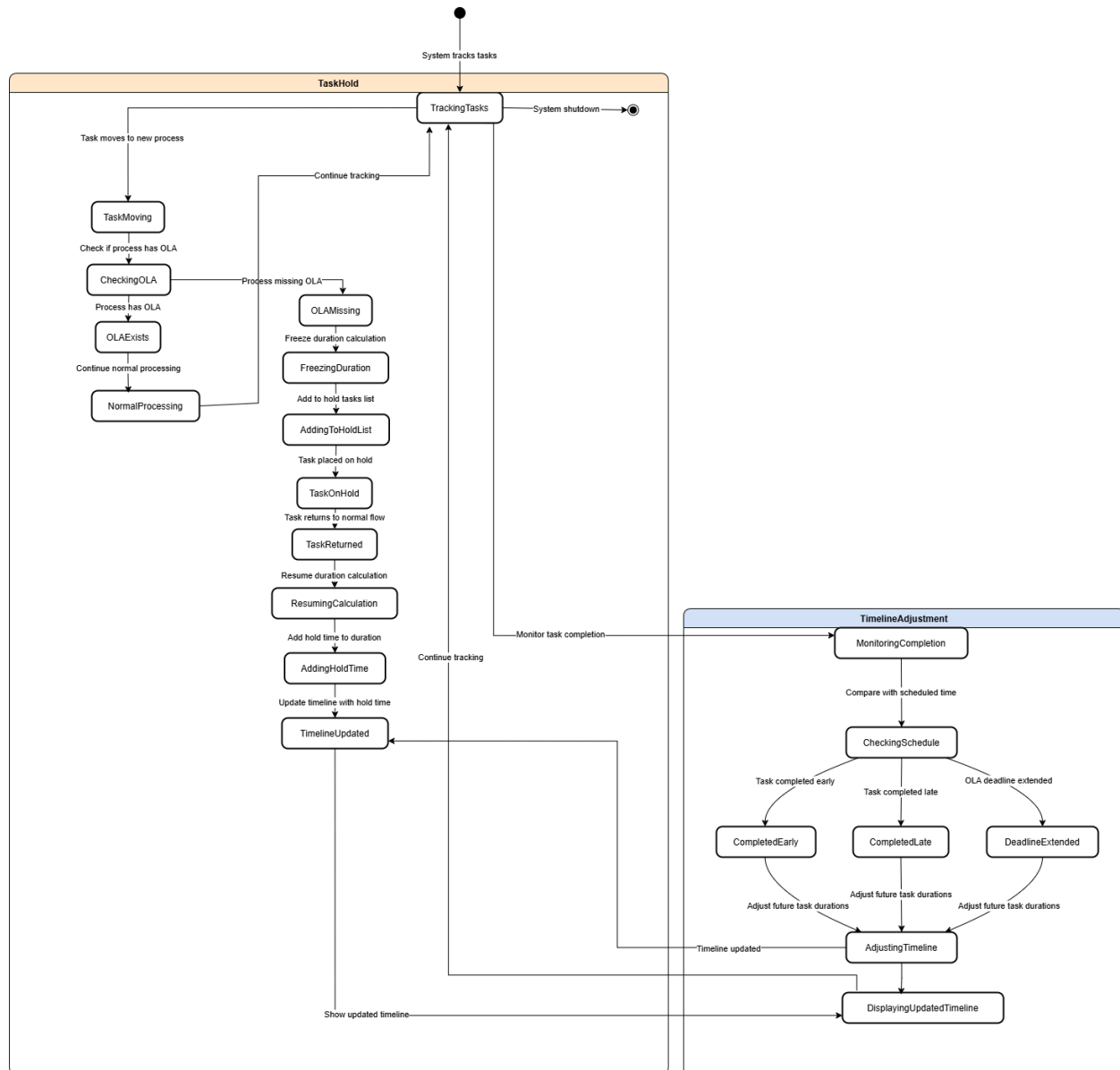
Use case 14: Mark task as urgent



Actor	General Manager
Preconditions	The General Manager is logged into the system. The task is part of an ongoing Planned Event (PE).
Flow of Step	1. The General Manager navigates to the PE section. 2. Searches for and selects the specific task within a PE. 3. Click on the “Mark as Urgent” option. Confirms the urgency update.

	4. The system updates the task status to " Urgent " and notifies relevant stakeholders.
Postconditions	The task is marked as urgent. Assigned users and workgroups receive notifications.

Use case 15: Update the estimated duration of an assigned task



Actor	Engineer-NET-PROJ-ACC-CABLE
Preconditions	The engineer is logged into the system.

	The task is assigned to the engineer. The task is in progress or pending an update.
Flow of Step	1. The engineer navigates to the assigned task. 2. Click on the “Update Estimated Duration” option. 3. Modifies the estimated duration. 4. Save the update. 5. The system records the new estimated duration.
Postconditions	The task’s estimated duration is updated. The system logs the change for tracking.

6. Use Case Diagram

